

# CABBY

Online Gaming Portal

## Product Requirements Document

Complete specification for the CABBY online gaming portal

## 1. Project overview

CABBY is a browser-based online gaming portal where users can discover and play free HTML5 games without downloading anything. Users visit the site, browse a grid of games sorted by category, click one, and play it directly inside the browser via an iframe.

The portal includes four engagement features beyond standard browsing: a daily missions system that gives all users the same 3 missions per calendar day, a streak counter that rewards consecutive days of completing all 3 missions, a pause bookmarks system that lets users save where they left off in any game with a personal note, and an automatic highest score tracker that records each user's best score per game.

The portal is a frontend-only project. There is no backend server, no database, no Firebase, and no API keys. All user data is stored in the browser using localStorage. Games are not built by the developer. They are third-party HTML5 games embedded via iframe.

### 1.1 Project at a glance

|                     |                                                                     |
|---------------------|---------------------------------------------------------------------|
| Project name        | CABBY: Online Gaming Portal                                         |
| Project type        | Browser-based online gaming portal                                  |
| Platform            | Browser, desktop and tablet                                         |
| Tech stack          | HTML, CSS, JavaScript. React acceptable but not required            |
| Backend             | None. No server, no database, no API keys                           |
| Data storage        | Browser localStorage only                                           |
| Engagement features | Daily missions, streak counter, pause bookmarks, best score tracker |
| Hosting             | GitHub Pages or Vercel free tier                                    |
| Deliverables        | Source code on GitHub, live deployed link, README                   |

## 2. Assets provided

The following files accompany this document. Do not replace or modify these assets.

| File                | Description                                                                                       | Usage                                                                                |
|---------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| logo.png            | CABBY brand logo. Yellow steering-wheel icon with CABBY text on a dark background                 | Place in the top-left corner of the navigation bar on all pages                      |
| reference_image.png | Portal layout reference showing the navigation bar, sidebar, missions section, and game card grid | Use as the visual target for overall layout, card design, and mission card placement |

## 3. Deliverables

The freelancer must deliver three items at the end of the project.

| # | Deliverable | Format                     | Notes                                                   |
|---|-------------|----------------------------|---------------------------------------------------------|
| 1 | Source code | Public GitHub repository   | All source files plus README.md with setup instructions |
| 2 | Live site   | Public URL                 | Deployed on GitHub Pages or Vercel free tier            |
| 3 | README.md   | Markdown file in repo root | Must include tech stack, setup steps, and live link     |

### Ownership

All source code, assets created during the project, and deployment rights transfer to the client on final payment. The freelancer may not resell or reuse any part of this project.

## 4. Website pages

The portal has exactly four pages. All pages share a top navigation bar that includes the streak counter. The Home, Leaderboard, and Profile pages also share a left sidebar.

| #   | Page             | Purpose                                                                             |
|-----|------------------|-------------------------------------------------------------------------------------|
| 4.1 | Home page        | Today's missions, browse and search games, filter by category                       |
| 4.2 | Game page        | Play the selected game inside an iframe. Save scores, bookmarks, view personal best |
| 4.3 | Leaderboard page | Tab switcher between All Scores and Your Best. Clear all option                     |
| 4.4 | Profile page     | Display name, four stat cards, favourites, bookmarks, reset                         |

### 4.1 Home page

The main landing page. Contains: top navigation bar with streak counter, left sidebar, Today's Missions section, and the game card grid.

#### Top navigation bar

| Element             | Position      | Behaviour                                                          |
|---------------------|---------------|--------------------------------------------------------------------|
| Logo and brand text | Left          | Logo image at 28x28px. Text CABBY in #F5C842, 16px, weight 500     |
| Search bar          | Centre        | Placeholder "Find game or genre". Filters grid in real time        |
| Streak counter      | Right section | Flame icon in #EF9F27 followed by current streak number in #F5C842 |
| Credits icon        | Right section | Bolt icon in #1D9E75 next to 0. Static display                     |
| Notification bell   | Right section | Bell icon with red badge. Static placeholder                       |
| User icon           | Right section | Person icon in #999                                                |
| Log in button       | Far right     | Background #1D9E75. Opens modal to set display name                |

#### Today's Missions section

Positioned above the game grid. Contains a heading "Today's Missions" and exactly 3 mission cards in a row. Each mission card shows the mission text and a completion indicator. Missions auto-complete. Completed cards use a green-tinted background (#1D3328) and show a green checkmark.

#### Left sidebar

A 48px wide vertical strip. Contains icon buttons stacked vertically, each 34x34px. Categories: All Games, Trending, Favourites, then a dot separator, then Racing, Puzzle, Action, Runner, Sports, New, Settings. Active icon uses background #3A3A3A and colour #F5C842.

### Game card grid

Minimum 9 game cards on load. Grid uses 3 columns at 1024px and wider, 2 columns at 768px to 1023px. Gap between cards: 10px.

| Card element  | Specification                                                    |
|---------------|------------------------------------------------------------------|
| Aspect ratio  | 16:10                                                            |
| Background    | Dark gradient, unique per genre                                  |
| Category icon | Centred, 32px, colour matching the genre                         |
| Game name     | Below icon, 12px, #DDDDDD, weight 500                            |
| Rating badge  | Bottom-left. Background #1D9E75, number in white 10px weight 500 |
| Bolt icon     | Top-left, #1D9E75                                                |
| Info button   | Top-right, 18px circle with info icon                            |
| Age tag       | Bottom-right, "16+" in #999 at 9px. Only on applicable games     |

## 4.2 Game page

Opens when a user clicks a game card. Contains the top nav bar with streak counter, no sidebar.

| Element                  | Specification                                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Back button              | Left arrow icon and "Back to games" text. Returns to Home page                                                                                              |
| Game title               | Displayed above the iframe. 18px, white, weight 500                                                                                                         |
| Best score display       | Above the iframe in the format "Your best: N" where N is read from cabby_best_scores. Shows "Your best: none" when no best exists for the game              |
| Iframe                   | Fills remaining width, minimum height 400px. Loads the game URL. Sandbox: allow-scripts allow-same-origin                                                   |
| Save Score input         | Below iframe. Number input. Placeholder "Enter your score"                                                                                                  |
| Save Score button        | Background #1D9E75. Saves {game, score, date} to cabby_scores. Also updates cabby_best_scores for the game ID only if the new score exceeds the stored best |
| Save Bookmark button     | Opens an inline text input that accepts up to 80 characters with a Save action. Stores {id, gameId, gameName, note, date, lastScore} to cabby_bookmarks     |
| Add to Favourites button | Heart icon toggle. Adds the game ID to cabby_favourites on first click, removes it on the next click                                                        |
| Error fallback           | If iframe fails to load: display "Game could not be loaded. Try another one."                                                                               |

## 4.3 Leaderboard page

Displays saved scores with two views. Contains the top nav bar with streak counter and left sidebar.

| Element    | Specification                |
|------------|------------------------------|
| Page title | Leaderboard with trophy icon |

| Element                 | Specification                                                                                                                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tab switcher            | Two tabs labelled "All Scores" and "Your Best". Switching tabs changes the visible table without reloading the page. The active tab is visually distinct                            |
| All Scores table        | 4 columns: Rank, Game Name, Score, Date. Sorted by Score descending. Data from <code>cabby_scores</code>                                                                            |
| Your Best table         | 4 columns: Rank, Game Name, Highest Score, Date achieved. One row per game with a recorded best score. Sorted by Highest Score descending. Data from <code>cabby_best_scores</code> |
| Empty state             | When the active tab has no data, show "No scores yet" or "No best scores yet"                                                                                                       |
| Clear All Scores button | Background #E24B4A. Confirmation prompt before deleting <code>cabby_scores</code> . Best scores are preserved                                                                       |

## 4.4 Profile page

Shows the user's local profile. Contains the top nav bar with streak counter and left sidebar.

| Element                  | Specification                                                                                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Avatar                   | 44px circle. User initials in white on #1D9E75 background. If no name: "?"                                                                                                |
| Display name             | 18px, weight 500, colour #FFFFFF. If not set: "Guest"                                                                                                                     |
| Edit Name button         | Opens inline text input. Saves to <code>cabby_username</code>                                                                                                             |
| Stat: Games Played       | Count of unique game names in <code>cabby_scores</code>                                                                                                                   |
| Stat: Total Scores Saved | Total entries in <code>cabby_scores</code> array                                                                                                                          |
| Stat: Favourites         | Count of items in <code>cabby_favourites</code> array                                                                                                                     |
| Stat: Best Streak        | Value of <code>cabby_streak.best</code> , with current value shown beneath                                                                                                |
| Favourites list          | All favourited game names with a remove button per item                                                                                                                   |
| Bookmarks list           | All saved bookmarks from <code>cabby_bookmarks</code> showing game name, note, and date. Each item has a delete button. Clicking the game name navigates to the Game page |
| Reset Profile button     | Background #E24B4A. Confirmation prompt. Clears all <code>cabby_keys</code>                                                                                               |

## 5. Daily missions and streaks

CABBY uses a daily missions system to encourage repeat visits. Each calendar day, all users see the same 3 missions chosen deterministically from a fixed pool of 10. Completing all 3 missions in a single day increments the user's streak. Missing a day resets the current streak.

## 5.1 Mission pool

| #  | Mission template                        |
|----|-----------------------------------------|
| 1  | Play any game for 2 minutes total       |
| 2  | Score above 500 in any game             |
| 3  | Try a game you have never played before |
| 4  | Play games from 3 different categories  |
| 5  | Add a game to favourites                |
| 6  | Save 3 scores in one day                |
| 7  | Set or change your display name         |
| 8  | Play a game for 5 minutes total         |
| 9  | Score above 1000 in any game            |
| 10 | Visit the leaderboard page              |

## 5.2 Mission selection and streak logic

| Rule                       | Specification                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------|
| Missions per day           | Exactly 3                                                                                                   |
| Determinism                | Same 3 missions for every user on the same calendar date                                                    |
| Selection formula          | Any deterministic function of date yielding 3 distinct missions from the pool                               |
| Reset time                 | Local midnight in the user's browser timezone                                                               |
| Auto-complete              | Mission checks fire whenever the relevant user action occurs                                                |
| Mission card height        | Approximately 80px                                                                                          |
| Mission card border radius | 8px                                                                                                         |
| Mission card grid          | 3 columns at viewport widths above 768px, 10px gap                                                          |
| Completed visual           | Background #1D3328, green checkmark icon in #1D9E75                                                         |
| Streak increment           | When all 3 of today's missions are complete, <code>cabby_streak.current</code> increases by 1               |
| Best streak update         | <code>cabby_streak.best = max(best, current)</code> on every increment, never reset                         |
| Streak reset               | On page load, if today's date is more than 1 day after <code>lastCompletedDate</code> , current resets to 0 |
| Nav display                | Flame icon in #EF9F27 + current value in #F5C842, on every page                                             |

## 6. Pause bookmarks

Pause bookmarks let users mark a moment in any game and return to it later with a personal note. Bookmarks are stored locally and managed from the Profile page.

| Rule              | Specification                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------|
| Trigger           | Save Bookmark button on the Game page, placed next to the Save Score button                                                  |
| Input             | Clicking opens an inline text input that accepts up to 80 characters with a Save action                                      |
| Storage key       | cabby_bookmarks (JSON array of bookmark objects)                                                                             |
| Bookmark fields   | id (unique string), gameId, gameName, note, date (YYYY-MM-DD), lastScore (the user's last entered score for that game, or 0) |
| Display location  | Bookmarks section on the Profile page                                                                                        |
| List item content | Game name, note, date, and a delete button                                                                                   |
| Delete action     | Removes that bookmark object from cabby_bookmarks                                                                            |
| Navigation        | Clicking the game name of any bookmark navigates to the Game page for that bookmarked game                                   |

## 7. Highest score tracker

Each user's highest score per game is tracked automatically, separate from the score history. Best scores are displayed on the Game page and on a dedicated Leaderboard tab.

| Rule              | Specification                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------|
| Storage key       | cabby_best_scores (JSON object mapping game ID to highest score number)                                     |
| Update trigger    | On every Save Score action on the Game page                                                                 |
| Update condition  | Update the value for the game ID only if the new score exceeds the stored best                              |
| Game page display | Shown above the iframe in the format "Your best: N", reading from cabby_best_scores for the current game ID |
| No-best fallback  | When no best is recorded for the current game ID, display "Your best: none"                                 |
| Leaderboard tab   | A "Your Best" tab on the Leaderboard page                                                                   |
| Your Best table   | Columns: Rank, Game Name, Highest Score, Date achieved. One row per game with a recorded best               |
| Sort order        | Highest Score descending. Highest goes to Rank 1                                                            |

## 8. Game data

Games are not built by the developer. They are defined as a static JavaScript array or JSON file. Each game object has the following fields.

| Field | Type   | Example       |
|-------|--------|---------------|
| id    | string | "speed-racer" |

| Field       | Type           | Example                          |
|-------------|----------------|----------------------------------|
| name        | string         | "Speed Racer"                    |
| genre       | string         | "racing"                         |
| rating      | number         | 72                               |
| ageTag      | string or null | "16+" or null                    |
| url         | string         | "https://example.com/game-embed" |
| description | string         | "A fast-paced racing game"       |

The developer must include at least 9 game entries. Game URLs can be placeholder URLs during development as long as the iframe loading mechanism works correctly.

9. localStorage specification

All keys must be prefixed with "cabby\_" to avoid conflicts with other sites.

| Key                  | Value type  | Description                                                              |
|----------------------|-------------|--------------------------------------------------------------------------|
| cabby_username       | string      | Display name set by the user                                             |
| cabby_scores         | JSON array  | Array of objects: {game, score, date}                                    |
| cabby_favourites     | JSON array  | Array of game ID strings                                                 |
| cabby_missions_state | JSON object | {date: "YYYY-MM-DD", completed: [bool, bool, bool]}                      |
| cabby_streak         | JSON object | {current, best, lastCompletedDate}                                       |
| cabby_played_games   | JSON array  | Array of game ID strings the user has played                             |
| cabby_session_times  | JSON object | Map of game ID to total seconds played                                   |
| cabby_bookmarks      | JSON array  | Array of bookmark objects: {id, gameId, gameName, note, date, lastScore} |
| cabby_best_scores    | JSON object | Map of game ID to the highest score recorded for that game               |

10. Visual design

The design must follow the layout shown in reference\_image.png. The portal uses a dark theme throughout.



## 10.1 Colour palette

| Colour             | Hex code | Used for                                                            |
|--------------------|----------|---------------------------------------------------------------------|
| Logo yellow        | #F5C842  | CABBY text, active sidebar icon, streak number, best score label    |
| Primary green      | #1D9E75  | Action buttons, rating badges, avatar background, mission checkmark |
| Danger red         | #E24B4A  | Delete buttons, notification badge                                  |
| Flame orange       | #EF9F27  | Streak flame icon in the navigation bar                             |
| Nav background     | #2A2A2A  | Top navigation bar, mission card surface                            |
| Sidebar background | #252525  | Left sidebar                                                        |
| Page background    | #1E1E1E  | Main page background                                                |
| Mission completed  | #1D3328  | Background for completed mission cards                              |
| Border colour      | #333333  | Borders and separators                                              |
| Text primary       | #EEEEEE  | Headings and main text                                              |
| Text secondary     | #999999  | Placeholder text, muted labels                                      |
| Search bar bg      | #3A3A3A  | Search input and active sidebar icon background                     |

## 10.2 Typography

| Element            | Size | Weight | Colour  |
|--------------------|------|--------|---------|
| Logo text          | 16px | 500    | #F5C842 |
| Page headings      | 15px | 500    | #EEEEEE |
| Mission card title | 12px | 500    | #DDDDDD |
| Game card name     | 12px | 500    | #DDDDDD |
| Rating badge       | 10px | 500    | #FFFFFF |
| Streak number      | 13px | 500    | #F5C842 |
| Best score label   | 13px | 500    | #F5C842 |
| Table text         | 13px | 400    | #CCCCCC |
| Stat card number   | 20px | 500    | #FFFFFF |
| Stat card label    | 12px | 400    | #999999 |

## 11. Technical requirements

| Requirement                   | Specification                                                    |
|-------------------------------|------------------------------------------------------------------|
| Tech stack                    | HTML, CSS, JavaScript. React is acceptable but not required      |
| Backend                       | None                                                             |
| Database                      | None. No Firebase, Supabase, MongoDB, or any external data store |
| API keys                      | None. The project must not require any API keys to run           |
| Data persistence              | localStorage only. All keys prefixed "cabby_"                    |
| Game embedding                | iframe with sandbox "allow-scripts allow-same-origin"            |
| Routing                       | Hash-based or JS page switching                                  |
| Page width range              | Responsive across 768px to 1440px wide                           |
| Responsive at 1024px+         | 3-column game grid, full sidebar visible                         |
| Responsive at 768px to 1023px | 2-column game grid, sidebar unchanged                            |
| Below 768px                   | Not required                                                     |
| Browser support               | Latest Chrome, Firefox, Edge                                     |
| Hosting                       | GitHub Pages or Vercel free tier. No paid hosting                |

## 12. Out of scope

### Important

The following items are not part of this project. Do not build these. Do not include code for these. If you think any of these are needed, contact the client before proceeding.

- Building actual games from scratch
- Multiplayer or co-op features
- Chat or social features
- Payment processing or subscriptions
- Mobile app versions (iOS or Android)
- Admin dashboard or CMS
- Ads or monetisation
- Third-party authentication
- Any feature requiring API keys or a backend
- Adding new missions beyond the 10 templates
- Custom mission editing by users
- Support for screens below 768px

### 13. Quality requirements

|                        |                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------|
| Visual fidelity        | The live site must match the layout shown in reference_image.png                                  |
| Logo placement         | The CABBY logo from logo.png must appear on every page in the navigation bar                      |
| Working localStorage   | All localStorage features must persist correctly across reloads                                   |
| Mission determinism    | All users must see the same 3 missions on the same calendar date                                  |
| Streak correctness     | Streak increments only on full mission completion and resets on missed days                       |
| Bookmark correctness   | Save, list, navigate, and delete actions all work correctly                                       |
| Best score correctness | Best scores update only when exceeded, display correctly on Game page and Your Best tab           |
| Navigation             | All four pages must be navigable without errors or broken links                                   |
| Final content          | No placeholder text such as "Lorem ipsum" in the final delivered site                             |
| Working README         | README.md must contain setup instructions that allow the client to run the project locally        |
| Revision rounds        | Up to 2 rounds of revisions after delivery. Changes outside this PRD require a separate agreement |

### 14. Developer choices and accepted options

The following decisions are left to the freelancer's judgement. Any option in each row is acceptable.

| Decision                  | Accepted options                                                                       |
|---------------------------|----------------------------------------------------------------------------------------|
| Framework                 | Vanilla JS, React, or any static-only stack                                            |
| Routing approach          | Hash-based routing, single-page JS switching, multi-page navigation                    |
| Build tool                | None, Vite, Create React App, any free static builder                                  |
| Hosting platform          | GitHub Pages, Vercel free tier                                                         |
| Icon library              | Any free icon library (Lucide, Tabler, Heroicons), inline SVG                          |
| Genre gradient palettes   | Developer chooses tones, must remain dark, genre-suggestive                            |
| Game data source          | Static array in JS file, static JSON file in repo                                      |
| Mission selection formula | Any deterministic function of date that yields 3 distinct missions from the pool of 10 |
| Bookmark ID generation    | Any unique string approach (timestamp, UUID, random hash)                              |

15. Tool requirements

| Category                | Specification                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------|
| Required tools          | Any code editor that saves UTF-8 plain text (VS Code, Sublime, Notepad++, etc.), Git         |
| Acceptable alternatives | Any text editor that can save UTF-8 plain text                                               |
| Forbidden               | No-code site builders, AI-generated full-site code generators that produce unreadable output |

16. Build and export specifications

| Requirement      | Specification                                                                        |
|------------------|--------------------------------------------------------------------------------------|
| Build output     | The live site must be deployable as static files only. No server runtime required    |
| Source files     | All source code committed to the GitHub repository in raw, editable form             |
| Production build | If using React, the production build output must also be deployable on a static host |

End of document

CABBY Product Requirements Document